

Future Software Architecture

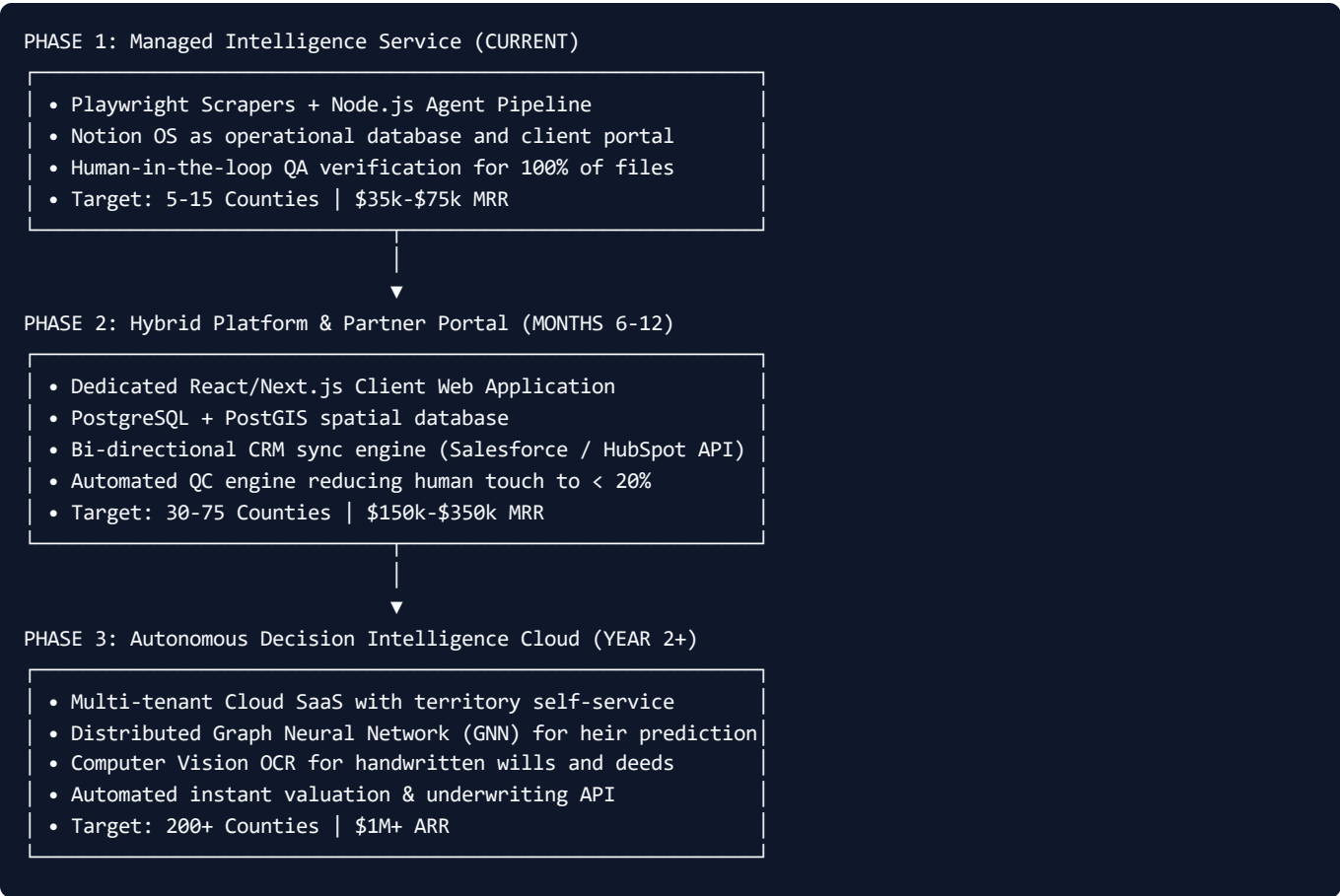
Technical roadmap and distributed cloud architecture transitioning Gieni from automated operator workflows into an institutional multi-tenant SaaS platform.



Platform Evolution Strategy: Service-to-Software Playbook

Gieni begins as a high-margin, county-exclusive managed intelligence platform powered by an autonomous agent pipeline and human-in-the-loop QA. As operational playbooks and entity resolution graphs achieve critical mass, the system transitions into a pure cloud-native Software-as-a-Service (SaaS) architecture. This roadmap outlines the target technical topology, microservices framework, and data isolation standards required for institutional enterprise scale.

1. Three-Phase Architecture Roadmap



2. Target Cloud Microservices Topology



1. Ingestion & Vision OCR Microservice

- Tech: Python / PyTorch / FastEmbed with AWS Textract & Gemini Vision.
- Responsibilities: Ingests unstructured court PDFs, scans historical deed books, extracts handwritten executor designations, and indexes text into structured JSON.



2. Entity Resolution & Knowledge Graph Service

- Tech: Neo4j Graph Database + Amazon Neptune.
- Responsibilities: Traverses complex relationships between deceased property owners, trust instruments, heir trees, and fiduciary appointments. Uncovers hidden decision-makers and resolves conflicts across multi-jurisdiction estates.



3. Valuation & Deal Friction Microservice

- Tech: FastAPI / LightGBM deployed on AWS SageMaker.
- Responsibilities: Computes real-time Net Equity Waterfalls, calculates the Deal Friction Score (DFS), and predicts transaction close probability based on historical county court speed.



4. Event Streaming & Multi-Channel Dispatch Gateway

- Tech: Apache Kafka / AWS EventBridge + Go worker fleet.
- Responsibilities: Sub-second routing of Priority A alerts, automated webhook delivery, and bi-directional synchronization with partner CRM environments.

3. Multi-Tenancy, Data Isolation & Security

Because Gieni guarantees county exclusivity, data isolation between partner accounts is mission-critical:

- **Row-Level Security (RLS):** PostgreSQL RLS policies guarantee that partner acquisition teams can only query opportunities assigned to their licensed county territories.
- **Zero Knowledge Territory Partitioning:** Client disposition notes, offer history, and revenue numbers are cryptographically isolated per tenant.
- **Regulatory Compliance & Privacy Vault:** Personal Identifiable Information (PII) is encrypted at rest using AES-256 with automated National DNC scrubbing.

4. Capital Efficiency & Engineering Milestones



Lean Software Evolution Principles

1. Build software only after manual procedures achieve 100% repeatability.
2. Fund engineering through county subscription cash flows rather than premature venture dilution.
3. Keep the customer experience focused on deals closed, not software complexity.